

AI-ASSISTED FULL STACK WEB DEVELOPMENT

Week 5 — Database Design & Management

1. Overview of the Week

Your app currently uses in-memory data — which disappears on every server restart. This week you add a real database. You will choose between PostgreSQL (via Supabase free tier) or MongoDB (via MongoDB Atlas free tier), design your schema, integrate an ORM/ODM, and migrate all your endpoints to use the database for full CRUD operations.

This week also includes the Month-1 Reflection Video – mandatory for all.

2. What You Will Do This Week

- **Choose your database:**
 - **PostgreSQL (via Supabase):** recommended if your data is structured with clear relationships (users, posts, comments).
 - **MongoDB (via Atlas):** recommended if your data is document-based with flexible schema (blog posts with variable fields, chat messages, etc.).
- **Set up the database:**
 - **Supabase:** go to supabase.com, create a free project, get the database URL from Project Settings → Database → Connection String.
 - **MongoDB Atlas:** go to mongodb.com/cloud/atlas, create a free cluster (M0 tier), whitelist your IP, create a database user, and get the connection string.
- **Design your schema/data model:** identify 2–3 entities (e.g., User, Task, Category), their fields, types, and relationships. Draw this as a simple diagram (Excalidraw, draw.io, or even hand-drawn + photo) and add it to your README.
- **Integrate an ORM or ODM:**
 - **PostgreSQL + Prisma:** npm install prisma @prisma/client, run npx prisma init, write schema in schema.prisma, run npx prisma migrate dev to create tables.
 - **MongoDB + Mongoose:** npm install mongoose, create model files with schemas, connect via mongoose in your server.
- **Refactor all 6+ endpoints** from Week 4 to use the database instead of in-memory data.
- **Verify full CRUD end-to-end:** open your frontend, create a record, see it appear, refresh the page — the data should persist. Update the record, delete it, etc.
- **Store environment variables safely:** database URL in .env, never committed.

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- **Record Month-1 Reflection Video** (5–7 min): what you built in Weeks 1–4, one big technical challenge and how you solved it, goals for Weeks 5–9. Upload to YouTube as Unlisted. Submit the link via the Deliverables Submission form.

3. Learning Outcomes for the Week

Week Outcome	Mapped to CLO
Design a database schema with clear entities and relationships	CLO-3
Integrate an ORM (Prisma) or ODM (Mongoose) with a cloud-hosted database	CLO-3
Refactor API endpoints to use persistent database storage	CLO-2, CLO-3
Complete mid-program reflection on technical progress	CLO-8

4. Resources

4.1 YouTube Videos

#	Title / Content	Channel & Link
1	SQL Tutorial — Full Course for Beginners	freeCodeCamp — https://www.youtube.com/watch?v=HXV3zeQKqGY
2	MongoDB in 100 Seconds	Fireship — https://www.youtube.com/watch?v=-bt_y4Loofg
3	PostgreSQL in 100 Seconds	Fireship — https://www.youtube.com/watch?v=n2Fluyr3lbc
4	Supabase — Full Course for Beginners	freeCodeCamp — https://www.youtube.com/watch?v=zBZgdTb-dns
5	MongoDB Atlas Complete Setup Guide	Traversy Media — https://www.youtube.com/watch?v=084rfmWdo3c
6	Prisma ORM Crash Course	Web Dev Simplified — https://www.youtube.com/watch?v=RebA5J-rlwg
7	Mongoose Crash Course	Traversy Media — https://www.youtube.com/watch?v=DZBGEVgL2eE
8	Database Design Course	freeCodeCamp — https://www.youtube.com/watch?v=ztHopE5Wnpc

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9	SQL vs NoSQL Explained	Traversy Media — https://www.youtube.com/watch?v=ruz-vK8IesE
10	Environment Variables in Node.js	Web Dev Simplified — https://www.youtube.com/watch?v=17UVejOw3zA

4.2 Blogs and Articles

#	Title	Source & Link
1	Prisma Getting Started Guide	Prisma Docs — https://www.prisma.io/docs/getting-started
2	Mongoose Documentation	Mongoose — https://mongoosejs.com/docs/guide.html
3	Supabase Quickstart	Supabase Docs — https://supabase.com/docs/guides/getting-started
4	MongoDB Atlas — Get Started	MongoDB Docs — https://www.mongodb.com/docs/atlas/getting-started
5	Database Schema Design Tutorial	Vertabelo Blog — https://vertabelo.com/blog/database-design-tutorial
6	SQL vs NoSQL: Choosing the Right Database	MongoDB Blog — https://www.mongodb.com/nosql-explained/nosql-vs-sql
7	PostgreSQL Tutorial — Basics	PostgreSQL Tutorial — https://www.postgresqltutorial.com
8	Secure Your .env File and API Keys	freeCodeCamp News — https://www.freecodecamp.org/news/how-to-use-env-variables-in-node
9	Indexing Basics for Faster Queries	DigitalOcean Community — https://www.digitalocean.com/community/tutorials/sql-indexes
10	One-to-Many and Many-to-Many Relationships	Prisma Blog — https://www.prisma.io/docs/concepts/components/prisma-schema/relations

4.3 MOOCs / Courses

#	Course	Platform & Link
1	Databases and SQL for Data Science	Coursera (IBM) — https://www.coursera.org/learn/sql-data-science
2	MongoDB University — M001: MongoDB Basics	MongoDB University (Free) — https://learn.mongodb.com

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3	Full Stack Open — Part 4 & 7 (DB + Deployment)	University of Helsinki (Free) — https://fullstackopen.com/en
4	Relational Database Developer	freeCodeCamp (Free) — https://www.freecodecamp.org/learn/relational-database

5. Deliverables

Deliverable 1: Database Integration in GitHub Repo

- All 6+ API endpoints now read from and write to the real database.
- Schema file committed:
 - Prisma users: schema.prisma in the repo.
 - Mongoose users: model files in /models folder.
- .env.example in the repo showing DATABASE_URL (or MONGO_URI) as required variables (with placeholder values).
- README updated with:
 - Database choice and why.
 - Schema diagram (screenshot or embedded image).
 - "Set up the database" section in the README.

Deliverable 2: Schema Diagram

- Simple visual showing your entities, fields, and relationships.
- Tools: Excalidraw (excalidraw.com), draw.io, dbdiagram.io, or Figma.
- Embed in README as an image, and submit as a separate PDF/image file.
- File name: W5_SchemaDiagram_[InternID].pdf or .png

Deliverable 3: CRUD Verification Screenshots

- Single PDF with screenshots showing each operation working end-to-end from frontend:
 - Create: new record added
 - Read: list of records displayed
 - Update: record updated successfully
 - Delete: record removed
- For each screenshot, include a caption describing what the user did.
- File name: W5_CRUDVerification_[InternID].pdf

Deliverable 4: Month-1 Reflection Video

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- Duration: 5–7 minutes.
- Content: 3 specific things you built or learned in Weeks 1–4 (with a code snippet or screenshot shown on screen), one challenge you faced and how you debugged it, goals for Weeks 5–9.
- Upload as **YouTube Unlisted** (not Public, not Private).
- Submit the link in the **Reflection Video Link** in the **Week 5 Submission Form** by **Sunday 11:59 PM IST of Week 5**.

6. Submission Instruction

- All submissions must be completed through the designated Google Form shared for Week 5 deliverables. **Submissions through any other medium will not be considered.**
- In case of multiple files, it is mandatory to **compress all files into a single .zip folder** before uploading. Only one consolidated file should be submitted in the file upload section of the form.
- If an intern is unable to meet the submission deadline due to a genuine emergency, a formal email must be sent to **internbti@geu.ac.in** by Sunday of the same week. Requests without valid justification or **submissions after the deadline** will not be considered.